

Roll No.

TEC-301

B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
ELECTRONIC DEVICES AND CIRCUITS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the following terms : (CO1)

(i) Emitter Injection Efficiency

(ii) Base Transport Factor

(iii) Punch Through

OR

(b) Explain the terms I_{CEO} and I_{CBO} with respect to leakage current. A BJT has $\alpha = 0.97$ and $I_{CO} = 50$ nA and is operating at base current of $10 \mu A$, find I_C , I_E , I_{CEO} and I_{CBO} . (CO1)

2. (a) Determine the voltage gain, current gain, input impedance and output impedance for RC coupled common-emitter amplifier. (CO1)

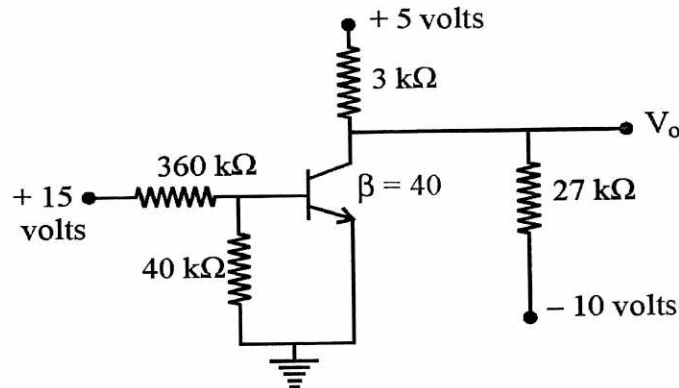
P. T. O.

(2)

OR

(b) Calculate V_o for the circuit shown below :

(CO1)



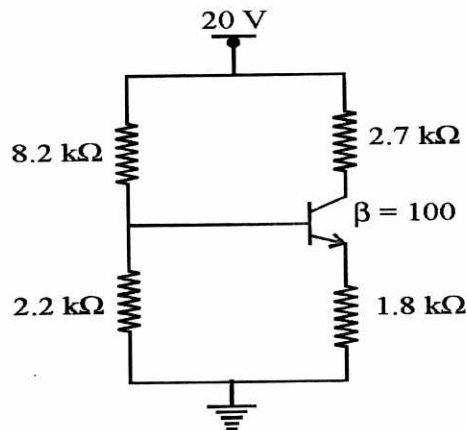
3. (a) Discuss short channel effect in MOSFET.

(CO1/CO2)

OR

(b) Calculate the stability factors (S) for the self-bias network shown below :

(CO1/CO2)



4. (a) Explain the input and output characteristics of CE configuration with proper diagrams.

(CO1/CO2)

OR

(b) Discuss the transfer and output characteristics of DE MOSFET.

(CO1/CO2)

(3)

5. (a) Explain the substrate bias effect in detail. Mention the methods by which the threshold voltage (V_{TH}) of MOSFET can be varied. (CO2)

OR

- (b) Discuss the MOS capacitor in detail, along with its CV characteristics. (CO2)

(3)

5. (a) Explain the substrate bias effect in detail. Mention the methods by which the threshold voltage (V_{TH}) of MOSFET can be varied. (CO2)

OR

- (b) Discuss the MOS capacitor in detail, along with its CV characteristics.

(CO2)

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TEC-302

**B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

DIGITAL ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Design the circuit of BCD Adder. (CO1)

OR

(b) Perform the following : (CO1)

(i) $(55)_8 - (37)_8$ using complement

(ii) $(26)_9 + (45)_9 = (\dots)$.

(iii) BCD Addition $972 + 885$

2. (a) Minimize using K-map : (CO1)

$$Y = \sum m (1, 5, 7, 13, 14, 15, 17, 18, 21, 22, 25, 29)$$

$$+ \sum d (6, 9, 19, 23, 30)$$

OR

(b) Minimize the expression using Quine McCluskey method : (CO1)

$$Y = \sum m (1, 3, 5, 7, 8, 11, 12, 13) + d (2, 6, 14)$$

P. T. O.

(2)

3. (a) Design the circuit for 2-bit magnitude comparator. (CO2)

OR

- (b) Design a circuit of 4 inputs which gives high output when only 2 inputs are high and low output when only 3 inputs are high. (CO2)

4. (a) Implement full Adder using multiplexer. (CO2)

OR

- (b) Design BCD to Excess 3 Code converter. (CO2)

5. (a) Implement the following expression using PLA and PAL : (CO2)

$$Y = \sum m(1, 2, 5, 7)$$

OR

- (b) What is the difference between Encoder and Decoder ? Design 5 : 32 using 3 : 8 decoder. (CO2)

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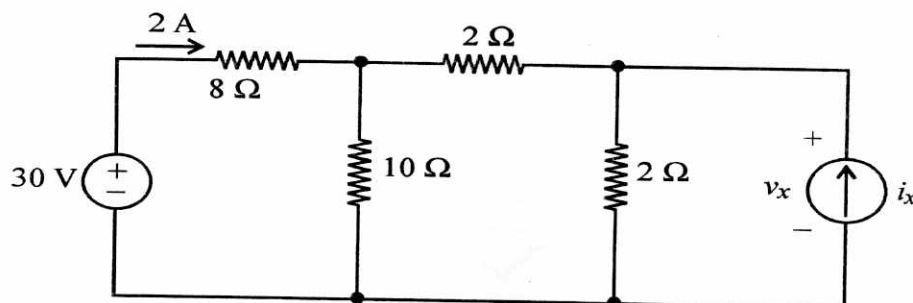
B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
NETWORK ANALYSIS AND SYNTHESIS

Time : 1½ Hours

Maximum Marks : 50

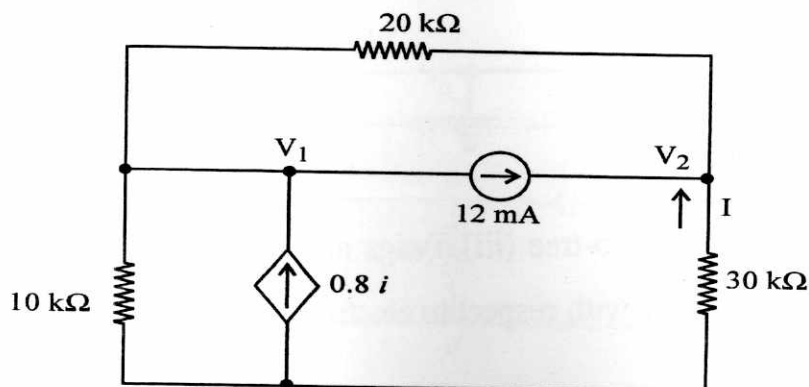
- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Determine v_x in the circuit shown below : (CO1)



OR

- (b) Calculate the node voltages in the circuit shown in figure below :



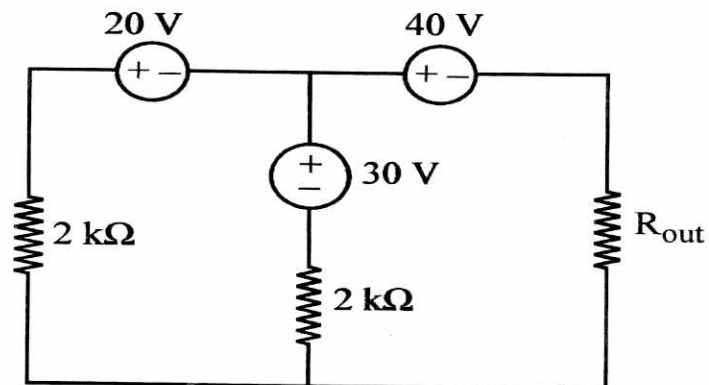
P. T. O.

2. (a) Consider the circuit shown in figure below :

(CO1)

(i) What is the maximum power that can be delivered to R_{out} ?

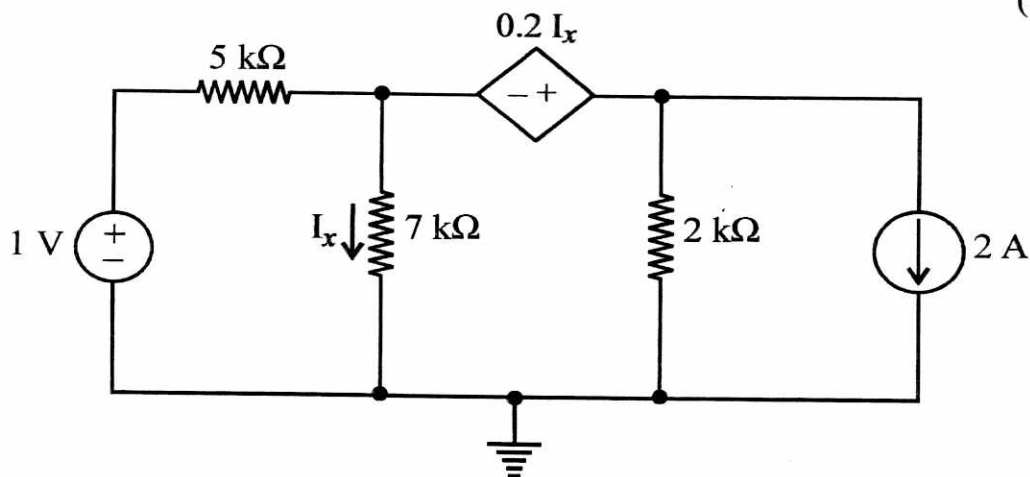
(ii) If $R_{out} = 3 \text{ k}\Omega$, find the power delivered to it.



OR

(b) Employ superposition principles to obtain a value for the current I_x .

(CO1)



3. (a) Define (i) Tree (ii) Co-tree (iii) Twigs and Links (iv) Tie-set schedule

(v) Cut-set schedule with respect to electric network.

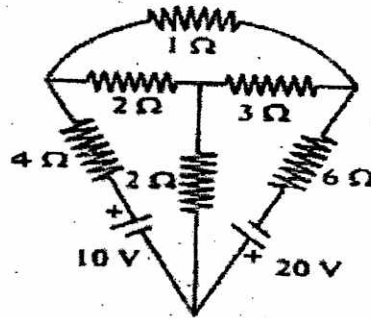
(CO2)

(3)

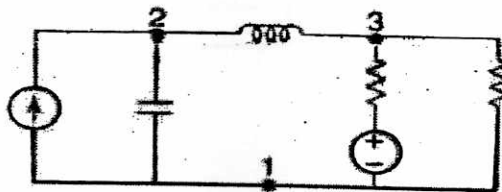
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OR

- (b) For the circuit shown below, write the equilibrium equations obtained from the Tie-set schedule. (CO2)

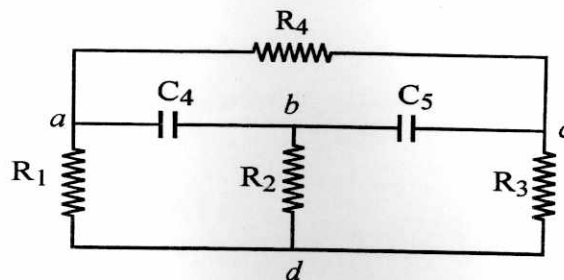


4. (a) How many trees are possible for the graph of the network of figure shown below and draw all possible trees : (CO2)



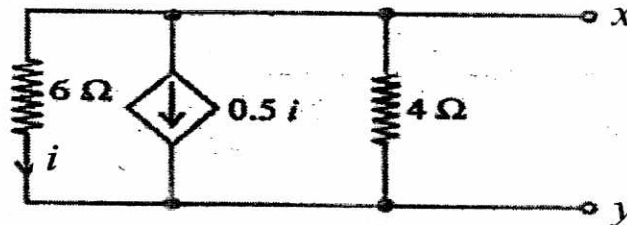
OR

- (b) Draw the directed graph of the circuit shown in figure below. Write the incidence matrix of the graph and determine the maximum number of possible trees of the graph. (CO2)



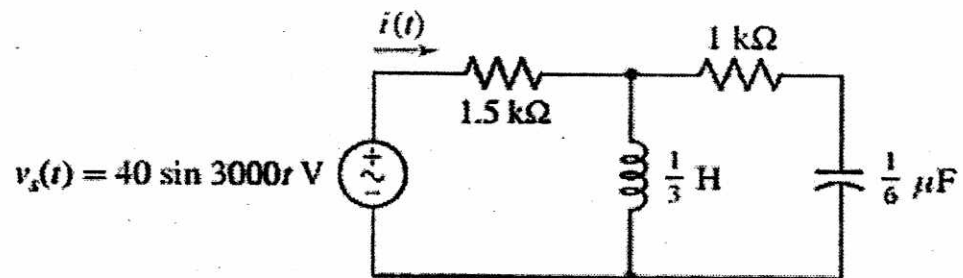
P. T. O.

5. (a) Find the Norton's equivalent circuit to the left of the terminal $x - y$ as shown in figure below : (CO1/CO2)



OR

- (b) Find the current $i(t)$ in the circuit shown in figure below : (CO1/CO2)



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TEC-304

B. TECH. (ECE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

SIGNALS AND SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Determine the even and odd components of the following signals :

(i) $x(t) = \dot{u}(t)$

(ii) $x[n] = u[n]$ (CO1)

OR

(b) Determine whether or not each of the following signals is periodic. If a signal is periodic, determine its fundamental period : (CO1)

(i) $x(t) = \cos(t) + \sin(\sqrt{2}t)$

(ii) $x[n] = e^{\frac{j2\pi n}{3}} + e^{\frac{j3\pi n}{4}}$

P. T. O.

2. (a) Determine whether the following signals are energy signals, power signals, or neither : (CO1)

(i) $x[n] = (-0.5)^n u[n]$

(ii) $x[n] = u[n]$

OR

- (b) Evaluate the following : (CO1)

(i) $\int_{-\infty}^{\infty} e^{-t} \delta(2t - 2) dt$

(ii) $\int_{-1}^1 (3t^2 + 1) \delta(t) dt$

(iii) $\int_1^2 (3t^2 + 1) \delta(t) dt$

3. (a) Determine the power and the rms value of the given signals : (CO1)

(i) $x(t) = 5 \cos\left(50t + \frac{2\pi}{3}\right)$

(ii) $x(t) = 10 \sin\left(50t + \frac{\pi}{4}\right) + 16 \cos\left(50t + \frac{\pi}{3}\right)$

OR

- (b) (i) Define discrete time unit step and discrete time unit impulse functions.

- (ii) Show that the product of two odd signals is an even signal. (CO1)

4. (a) Check whether the given system $y(t) = \int_{-\infty}^t x(\tau) d\tau$ is : (CO2)

- (i) Linear or nonlinear

- (ii) Time invariant or time variant

(3)

OR

- (b) Let $y(t) = x(t) * h(t)$. Then show that : (CO2)

$$x(t - t_1) * h(t - t_2) = y(t - t_1 - t_2)$$

5. (a) Consider a continuous time LTI system with the input-output relation given by $y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau) d\tau$. Find the impulse response $h(t)$ of this system. (CO2)

OR

- (b) What is LTI system and its impulse response ? Explain how the impulse response is useful in finding out the LTI system output due to any arbitrary input signal. (CO2)

(3)

OR

(b) Let $y(t) = x(t) * h(t)$. Then show that : (CO2)

$$x(t - t_1) * h(t - t_2) = y(t - t_1 - t_2)$$

5. (a) Consider a continuous time LTI system with the input-output relation given by $y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau) d\tau$. Find the impulse response $h(t)$ of this system. (CO2)

OR

(b) What is LTI system and its impulse response ? Explain how the impulse response is useful in finding out the LTI system output due to any arbitrary input signal. (CO2)

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TMA-310

B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
ADVANCED ENGINEERING MATHEMATICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Write down the statements of the following series/theorems : (CO1)

(i) Taylor series

(ii) Laurent series

(iii) Residues theorem

OR

(b) Show that the function : (CO1)

$$f(z) = u + iv$$

where $f(z) = e^{-z^{-4}}$ ($z \neq 0$) and $f(0) = 0$ is not analytic at $z = 0$
although Cauchy-Riemann equations are satisfied at origin.

P. T. O.

2. (a) Write down the statement of Cauchy's residues theorem, and hence to compute the value of the integral $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where $C: |z|=3$. (CO1)

OR

- (b) Define the analytic function, and hence to construct that analytic function $f(z)$ using Milne's Thomson method where the real part of function $f(z)$ is : (CO1)

$$u(x, y) = e^{-x} \left[(x^2 - y^2) \cos y + 2xy \sin y \right]$$

3. (a) Using Bisection method to compute a real root of the equation $x^3 - 5x + 3 = 0$ correct to three decimal places. Perform only eight iterations. (CO3)

OR

- (b) Define the harmonic function. Prove that : (CO1)

$$u = x^2 - y^2 - 2xy - 2x + 3y$$

is harmonic. Find a function v such that $f(z) = u + iv$ is analytic.

4. (a) Find all values of k such that the function $f(z) = e^x [\cos ky + i \sin ky]$ is analytic. (CO1)

OR

- (b) Evaluate the complex integral $\int_{1-i}^{2+3i} (z + z^2) dz$ along the line joining the points $(1, -1)$ and $(2, 3)$. (CO1)

(3)

5. (a) Newton-Raphson method is used to compute a root of the equation $x^3 - x - 1 = 0$ with initial condition $x_0 = 1.5$ (Correct to three decimal places). (CO3)

OR

- (b) Define the following with the help of *one* example : (CO2)
Translation, Magnification, Rotation and Inversion.